Machinery Lubrication

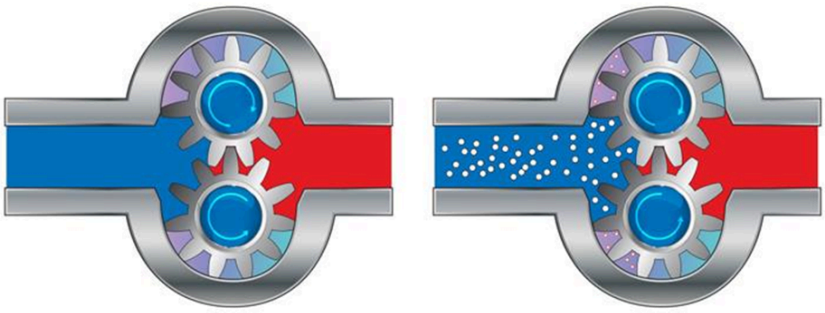
Published by



Share Post Share Subscribe

Troubleshooting Hydraulic Pumps

Al Smiley, GPM Hydraulic Consulting



When a hydraulic issue occurs, the pump is one of the first components changed out, but it should be the last. A pump is one of the most time-consuming and expensive parts to replace, and it should never be changed before several tests are conducted. In this article, we'll discuss some of the simplest tests and checks to perform first before you consider part replacement.

Visual Tests

Before any physical tests are administered, there are several, basic visual tests that you can conduct to ascertain the state of your pump.

Check that the electric motor is running. Although this is a simple concept, before you begin replacing parts, it's critical that you make sure the electric motor is running. This can often be one of the easiest aspects to overlook, but it is necessary to confirm before moving forward.

Check that the pump shaft is rotating. Even though coupling guards and C-face mounts can make this difficult to confirm, it is important to establish if your pump shaft is rotating. If it isn't, this could be an indication of a more severe issue, and this should be investigated immediately.

Check the oil level. This one tends to be the more obvious check, as it is often one of the only factors inspected before the pump is changed. The oil level should be three inches above the pump suction. Otherwise, a vortex can form in the reservoir, allowing air into the pump.

If the oil level is low, determine where the leak is in the system. Although this can be a difficult process, it is necessary to ensure your machines are performing properly. Leaks can be difficult to find.

Sound Checks

What does the pump sound like when it is operating normally? Vane pumps generally are quieter than piston and gear pumps. If the pump has a high-pitched whining sound, it most likely is cavitating. If it has a knocking sound, like marbles rattling around, then aeration is the likely cause.

Cavitation

Cavitation is the formation and collapse of air cavities in the liquid. When the pump cannot get the total volume of oil it needs, cavitation occurs. Hydraulic oil contains approximately nine percent dissolved air. When the pump does not receive adequate oil volume at its suction port, high vacuum pressure occurs.

This dissolved air is pulled out of the oil on the suction side and then collapses or implodes on the pressure side. The implosions produce a very steady, high-pitched sound. As the air bubbles collapse, the inside of the pump is damaged.

While cavitation is a devastating development, with proper preventative maintenance practices and a quality monitoring system, early detection and deterrence remain attainable goals. UE System's UltraTrak 850S CD pump cavitation sensor is a Smart Analog Sensor designed and optimized to detect cavitation on pumps earlier by measuring the ultrasound produced as cavitation starts to develop early-onset bubbles in the pump. By continuously monitoring the impact caused by cavitation, the system provides a simple, single value to trend and alert when cavitation is occurring.

3 Reasons Why a Pump Cavitates

Reason One

The oil viscosity is too high. Low oil temperature increases the oil viscosity, making it harder for the oil to reach the pump. Most hydraulic systems should not be started with the oil any colder than 40°F and should not be put under load until the oil is at least 70°F.

Many reservoirs do not have heaters, particularly in the South. Even when heaters are available, they are often disconnected. While the damage may not be immediate, if a pump is continually started up when the oil is too cold, the pump will fail prematurely.

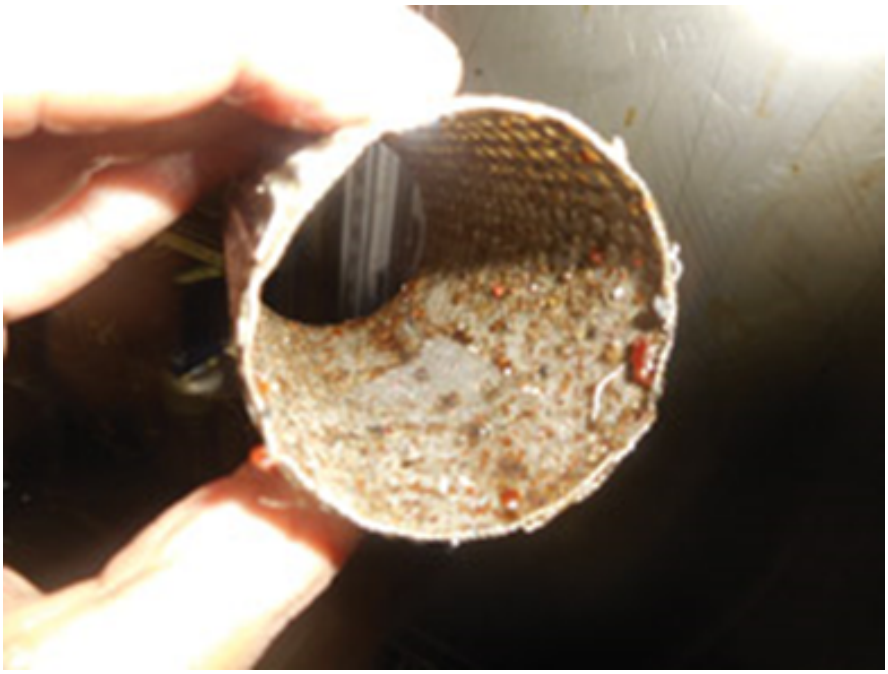


Figure 1

Reason Two

The suction filter or strainer is contaminated. A strainer is typically 74 or 149 microns in size and is used to keep “large” particles out of the pump. The strainer may be located inside or outside the reservoir. Strainers located inside the reservoir are out of sight and out of mind. Many times, maintenance personnel are not even aware that there is a strainer in the reservoir.

The suction strainer should be removed from the line or reservoir and cleaned a minimum of once a year. Years ago, a plant sought out help to troubleshoot a system that had already had five pumps changed within a single week. Upon closer inspection, it was discovered that the breather cap was missing, allowing dirty air to flow directly into the reservoir.

A check of the hydraulic schematic showed a strainer in the suction line inside the tank. When the strainer was removed, a shop rag was found wrapped around the screen mesh. Apparently, someone had used the rag to plug the breather cap opening, and it had then fallen into the tank. Contamination can come from a variety of different sources, so it pays to be vigilant and responsible with our practices and reliability measures.

Reason Three

The electric motor is driving the hydraulic pump at a speed that is higher than the pump’s rating. All pumps have a recommended maximum drive speed. If the speed is too high, a higher volume of oil will be needed at the suction port.

Due to the size of the suction port, adequate oil cannot fill the suction cavity in the pump, resulting in cavitation. Although this rarely happens, some pumps are rated at a maximum drive speed of 1,200 revolutions per minute (RPM), while others have a maximum speed of 3,600 RPM. The drive speed should be checked any time a pump is replaced with a different brand or model.

Every one of these devastating causes of cavitation threatens to cause major, irreversible damage to your equipment. Therefore, it’s not only critical to have proper, proactive practices in place, but also a monitoring system that can continuously protect your valuable assets, such as UE System’s UltraTrak 850S CD pump cavitation senor. These sensors regularly monitor the health of your pumps and alert you immediately if cavitation symptoms are present, allowing you to take corrective action before it’s too late.

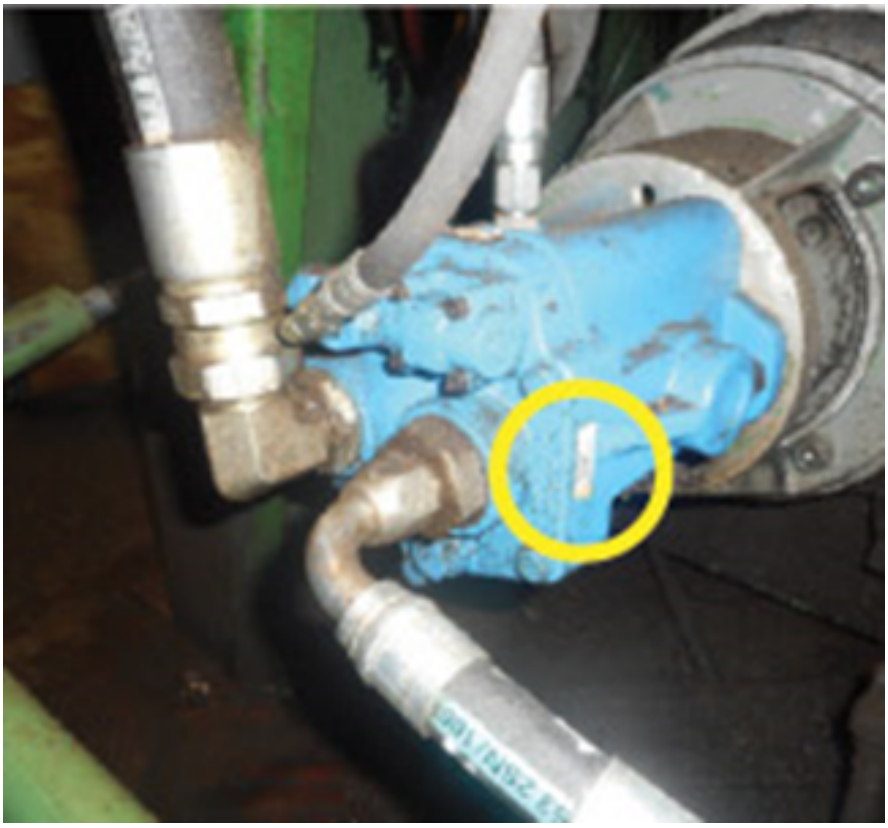


Figure 2. *An arrow on the gear pump’s housing indicates the direction of rotation.*

Aeration

Aeration is sometimes known as pseudo cavitation because air is entering the pump suction cavity. However, the causes of aeration are entirely different than that of cavitation. While cavitation pulls air out of the oil, aeration is the result of outside air entering the pump's suction line.

Several factors can cause aeration, including an air leak in the suction line. This could be in the form of a loose connection, a cracked line, or an improper fitting seal. One method of finding the leak is to squirt oil around the suction line fittings. The fluid will be momentarily drawn into the suction line, and the knocking sound inside the pump will stop for a short period of time once the airflow path is found.

A bad shaft seal can also cause aeration if the system is supplied by one or more fixed displacement pumps. Oil that bypasses inside a fixed displacement pump is ported back to the suction port. If the shaft seal is worn or damaged, air can flow through the seal and into the pump's suction cavity.

As mentioned previously, if the oil level is too low, oil can enter the suction line and flow into the pump. Therefore, always check the oil level with all cylinders in the retracted position.

If a new pump is installed and pressure will not build, the shaft may be rotating in the wrong direction. Some gear pumps can be rotated in either direction, but most have an arrow on the housing indicating the direction of rotation, as depicted in Figure 2.

Pump rotation should always be viewed from the shaft end. If the pump is rotated in the wrong direction, adequate fluid will not fill the suction port due to the pump's internal design.

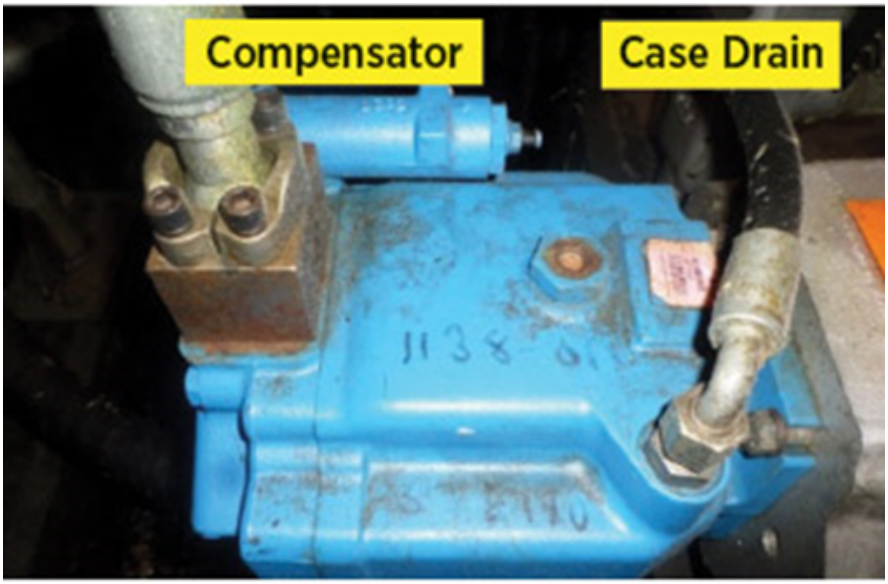


Figure 3. *The compensator setting limits the maximum pressure at the outlet port of a variable displacement pump.*

Testing a Fixed Displacement Pump

A fixed displacement pump delivers a constant volume of oil for a given shaft speed. A relief valve must be included downstream of the pump to limit the maximum pressure in the system.

After the visual and sound checks are made, the next step is to determine whether you have a volume or pressure problem. If the pressure will not build to the desired level, isolate the pump and relief valve from the system. This can be done by closing a valve, plugging the line downstream, or blocking the relief valve. If the pressure builds when this is done, there is a component downstream of the isolation point that is bypassing. If the pressure does not build up, the pump or relief valve is bad.

If the system is operating at a slower speed, a volume problem exists. Pumps wear over time, which results in less oil being delivered. While a flow meter can be installed in the pump's outlet line, this is not always practical, as the proper fittings and adapters may not be available. To determine if the pump is badly worn and bypassing, first check the current to the electric motor. If possible, this test should be made when the pump is new to establish a reference. Electric motor horsepower is relative to the hydraulic horsepower required by the system.

This is shown in the following formula:

electric motor horsepower (hp) = gallons per minute (GPM) x pounds per square inch (psi) x 0.00067.

For example, if a 50-GPM pump is used and the maximum pressure is 1,500 psi, a 50-hp motor will be required. If the pump is delivering less oil than when it was new, the current to drive the pump will drop. A 230-volt, 50-hp motor has an average full load rating of 130 amps. If the amperage is considerably lower, the pump is most likely bypassing and should be changed.

The temperature of the pump housing and suction line should also be checked. A severe increase in temperature indicates a badly worn pump.

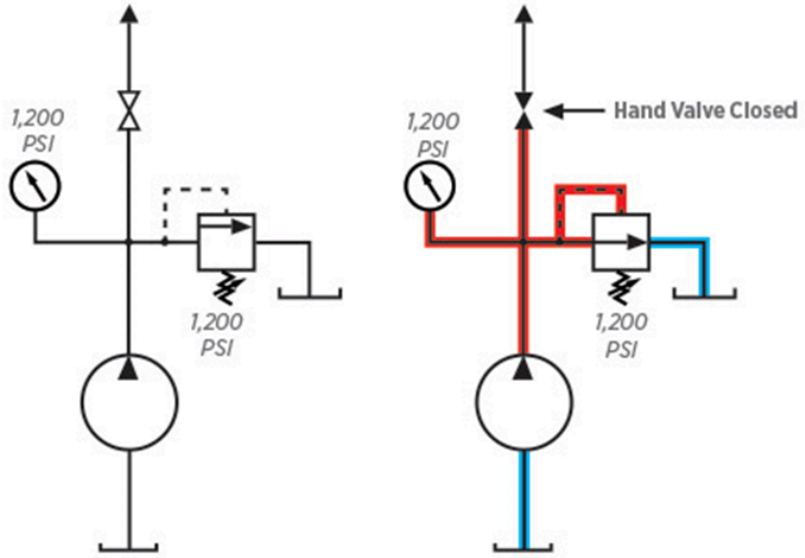


Figure 4. *To isolate a fixed displacement pump and relief valve from the system, close a valve or plug the line downstream* ***(left)****. If pressure builds, a component downstream of the isolation point is bypassing* ***(right)****.*

Testing a Variable Displacement Pump

The most common type of variable displacement pump is the pressure-compensating design. The compensator setting limits the maximum pressure at the pump's outlet port. The pump should be isolated as described for the fixed displacement pump.

If pressure does not build up, the relief valve or pump compensator may be bad. Prior to checking either component, perform the necessary lockout procedures and verify that the pressure at the outlet port is zero psi. The relief valve and compensator can then be taken apart and checked for contamination, wear, and broken springs.

If a volume problem exists in the system, perform the following tests:

1. Check the tank line temperature of the relief valve with a temperature gun or infrared camera. The tank line should be near ambient temperature. If the line is hot, the relief valve is either stuck partially open or is set too low.
2. Install a flow meter in the case drain line and check the flow rate. Most variable displacement pumps bypass one to three percent of the maximum pump volume through the case drain line. If the flow rate reaches 10 percent, the pump should be changed. Permanently installing a flow meter in the case drain line is an excellent reliability and troubleshooting tool.
3. Check the current on the drive motor.
4. Ensure the compensator is 200 psi above the maximum load pressure. If set too low, the compensator spool will shift and start reducing the pump volume when the system is calling for maximum volume.

Performing these recommended tests should help you make good decisions about the condition of your pumps or the cause of pump failures. If you change a pump, have a reason for changing it. Don't just do it because you have a spare one in stock.

Conduct a reliability assessment on each of your hydraulic systems so when an issue occurs, you will have current pressure and temperature readings to consult.



About the Author
Al Smiley
Al Smiley is the president of GPM Hydraulic Consulting Inc., located in Monroe, Georgia. Since 1994, GPM has provided hydraulic training, consulting and reliability assessments to companies in t... Read More

Machinery Lubrication (2/2018)

Additional Articles



Whitepaper: Improving Metalworking Fluid Performance and Service Life Through Cleanliness



Mobil 1™ Thermal Management Fluid Helps All-Electric Porsche GT4 e-Performance



How Water Causes Bearing Failure



Answers to Your Questions on Automotive Lubricants



Related Articles

- Utility's Cooling Tower: The Unsung Hero of the Lubrication Industry
- Unboxing: Mystik® JT-9™ LeakShield® AW Hydraulic Oil
- Why You Shouldn't Overlook Hydraulic Pipe Size
- 4 Steps to Diagnose Low System Pressure



Featured Videos

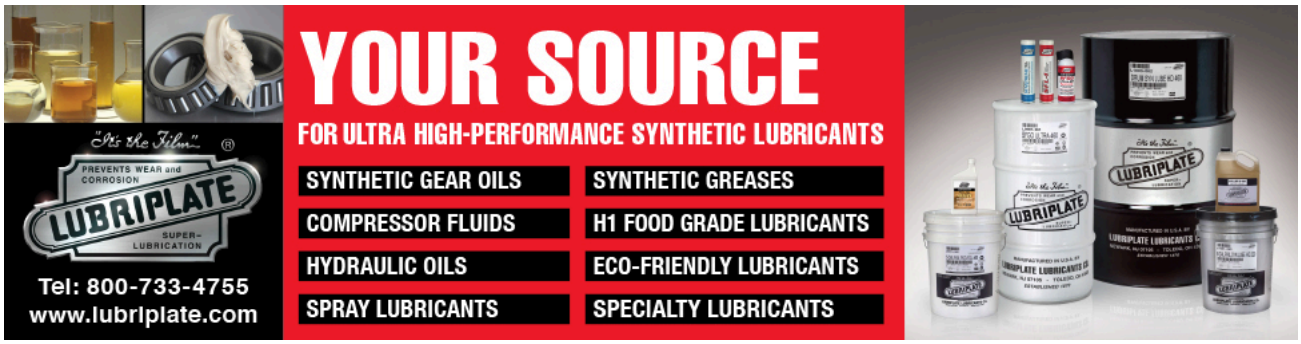
- Unboxing: Luneta Air-Lock
- How to Develop an Effective Maintenance Training Program
- Unboxing: perma STAR VARIO EX / STAR VARIO BLUETOOTH EX

Featured Whitepapers

- Permanently Installed Technology Improves Motor Reliability
- The Reliability Leader's Budget Playbook

Buyer's Guide

- Hydraulics
- Oil Sampling Hardware
- Filtration and Filtration Systems



Get More Info

Glossary
About Noria
Privacy Policy
Advertise With Us Submit Content

Contact Us

Send Us a Message
800-597-5460
Fax: 918-746-0925

Other Noria Sites

Reliable Plant
Reliable Plant Conference
Noria
Noria Bookstore
RelaWorks

1328 E. 43rd Court, Tulsa, OK 74105 · © 2026 Noria Corporation